Supplementary Information

Wormhole Formation in Matter-Coupled Vacuum Transitions: Gravitational Collapse, Horizon Constraints, and the Limits of Cosmological Rebirth

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S1. Equations and scheme

The metric is $ds^2=-f\,du\,dv+r^2d\Omega^2$ with $f=e^{\sigma}$, $G_{\mu\nu}=T_{\mu\nu}$ ($8\pi G=1$) and $T_{\mu\nu}=\partial_\mu\varphi\partial_\nu\varphi-g_{\mu\nu}[(\nabla\varphi)^2/2+U]$. A symbolic computation of the Einstein tensor (derive_equations.py) gives

$$r_{uu}-\sigma_u r_u=-\frac{r}{2}\phi_u^2,\qquad r_{uv}=-\frac{r_u r_v}{r}-\frac{f}{4r}+\frac{Ufr}{4} \tag{S1}$$

$$\phi_{uv}=-\frac{\phi_u r_v+\phi_v r_u}{r}-\frac{f}{4}U'(\phi),\qquad \sigma_{uv}=\frac{f+4r_u r_v}{2r^2}-\phi_u\phi_v \tag{S2}$$

The first of Eqs. (S1) is the constraint along u (that along v is Eq. (4) of the paper) and the others are the evolution equations. The scheme advances rows of constant u on a uniform lattice $u_j=jh$, $v_i=ih$. A cell with corners $W=(j,i-1)$, $A=(j+1,i-1)$, $B=(j,i)$ gives the new point $C=(j+1,i)$ by $X_C=X_A+X_B-X_W+h^2X_{uv}$, with X_{uv} evaluated at the cell centre from the mean values and the central differences, iterated three times (predictor–corrector). Near the centre the evolution of σ divides by a small radius and its rounding noise is amplified, so within $K=6$ points of the centre σ is integrated from the constraint $\sigma_v=(r_{vv}+r\varphi_v^2/2)/r_v$ (trapezoid), and beyond it from the evolution equation. Each row is solved with the centre value of σ as the unknown: elementary flatness requires $\sigma_c=\ln(4r_v^2)$ at the centre, which defines a residual that is solved by the secant method (ten passes at most); an iteration on the plain residual has a gain of modulus greater than one and diverges. The centre value of φ follows from $\varphi_t=2\varphi_v$ at the centre by an implicit trapezoid with the one-sided derivative (the explicit iteration has gain -1.5).

The data on the initial cone are $r=v/2$, $\varphi(0,v)=A\psi(v)$ and $\sigma(0,v)=\mathrm{int}_0^v\, r\varphi_v^2dv$, the integral of the constraint along v with $f=1$ at the centre. The potential $U=(\kappa/4)(\psi^2-1)^2+s\psi+d$ is tuned by a root finder so that the false minimum has density 3 and the true minimum $-3q$.

S2. Numerical diagnostics

Misner–Sharp mass of exact solutions: Table 3 of the paper. The diagnostics of each run are the signs of r_u (a forward difference in u) and r_v (a central difference in v) at every point more than four cells from the centre and eight from the outer boundary; the quadrant map of Fig. 4 is produced by the same classification. The residuals of the constraints are $C_{vv}=r_{vv}-\sigma_v r_v+(r/2)\varphi_v^2$ and $C_{uu}=r_{uu}-\sigma_u r_u+(r/2)\varphi_u^2$, with second differences and central first differences on the lattice.

S3. Thin-wall check

Table S0 lists the check of the critical mass. For each of nine combinations of q and $\hat{o}$ the finite-tension value $m_c=\max_R m_+(R)$ is computed on a grid of two million points, and the existence of a forbidden window is tested at $m_c(1\mp10^{-3})$.

q	$\hat{\sigma}$	m_c	Window at $m_c(1-10^{-3})$	Window at $m_c(1+10^{-3})$
0.5	0.01	1.49999	yes	no
0.5	0.30	1.48757	yes	no
0.5	0.90	1.39332	yes	no
1.0	0.01	1.99998	yes	no
1.0	0.30	1.98505	yes	no
1.0	0.90	1.86927	yes	no
2.0	0.01	2.99998	yes	no
2.0	0.30	2.98253	yes	no
2.0	0.90	2.84482	yes	no

Table S0. Check of the critical mass (units of R_{dS}).

S4. Scan of the collapse runs

Supplementary Table S1 lists all runs: field amplitude A, wall width w, core radius $R_w=V_w/2$, exterior mass m_{ext} (standard deviation across the exterior in parentheses), $R_M=(3m_{ext}/\Delta)^{1/3}$, the radius of the first trapped point at the base resolution, the horizon radius r_h of the measured mass, and the number of anti-trapped points in the core on the last row.

A	w	R_w	m_{ext}	R_M	r (first trapped)	r_h	Anti-trapped core points
0.05	0.05	0.40	0.166 (0.000)	0.437	0.163	0.162	0
0.05	0.05	0.60	0.514 (0.000)	0.636	0.429	0.433	0
0.05	0.05	0.80	1.158 (0.000)	0.834	0.743	0.745	0
0.05	0.05	1.00	2.188 (0.001)	1.030	1.044	1.045	0
0.05	0.05	1.20	3.687 (0.001)	1.226	1.330	1.331	31
0.05	0.05	1.50	7.001 (0.001)	1.518	1.738	1.739	127
0.05	0.05	2.00	16.033 (0.003)	2.001	2.387	2.390	287
0.05	0.05	3.00	51.525 (0.007)	2.953	3.627	3.632	607
0.05	0.10	0.40	0.198 (0.000)	0.462	—	0.191	0
0.05	0.10	0.60	0.589 (0.001)	0.665	0.474	0.479	0
0.05	0.10	0.80	1.301 (0.001)	0.867	0.791	0.796	0
0.05	0.10	1.00	2.426 (0.002)	1.066	1.095	1.099	0
0.05	0.10	1.20	4.054 (0.003)	1.266	1.382	1.387	0
0.05	0.10	1.50	7.638 (0.006)	1.563	1.795	1.801	59
0.05	0.10	2.00	17.393 (0.011)	2.056	2.455	2.462	159
0.05	0.10	3.00	55.941 (0.030)	3.036	3.725	3.737	359
0.05	0.20	0.40	0.324 (0.001)	0.545	—	0.298	0
0.05	0.20	0.60	0.932 (0.003)	0.775	0.626	0.653	0
0.05	0.20	0.80	2.000 (0.006)	1.000	0.968	1.000	0
0.05	0.20	1.00	3.656 (0.009)	1.223	1.285	1.326	0
0.05	0.20	1.20	6.042 (0.013)	1.446	1.590	1.639	0
0.05	0.20	1.50	11.381 (0.021)	1.785	2.039	2.101	19
0.05	0.20	2.00	48.383 (4.594)	2.892	—	3.552	119
0.05	0.20	3.00	110.863 (8.554)	3.813	—	4.735	319
0.10	0.05	0.40	0.191 (0.000)	0.457	0.183	0.185	0
0.10	0.05	0.60	0.554 (0.000)	0.652	0.456	0.458	0
0.10	0.05	0.80	1.194 (0.000)	0.842	0.755	0.758	0
0.10	0.05	1.00	2.179 (0.000)	1.029	1.041	1.043	0
0.10	0.05	1.20	3.569 (0.001)	1.213	1.310	1.312	31
0.10	0.05	1.50	6.535 (0.001)	1.484	1.690	1.692	127
0.10	0.05	2.00	14.238 (0.002)	1.924	2.285	2.286	287
0.10	0.05	3.00	42.413 (0.007)	2.768	3.389	3.392	607
0.10	0.10	0.40	0.203 (0.000)	0.467	—	0.196	0
0.10	0.10	0.60	0.592 (0.001)	0.666	0.480	0.481	0
0.10	0.10	0.80	1.285 (0.001)	0.863	0.789	0.791	0
0.10	0.10	1.00	2.362 (0.002)	1.057	1.084	1.085	0
0.10	0.10	1.20	3.900 (0.003)	1.249	1.362	1.364	0

A	w	R_w	m_{ext}	R_M	r (first trapped)	r_h	Anti-trapped core points
0.10	0.10	1.50	7.226 (0.006)	1.534	1.760	1.761	59
0.10	0.10	2.00	16.052 (0.011)	2.002	2.389	2.391	159
0.10	0.10	3.00	49.421 (0.030)	2.913	3.575	3.579	359
0.10	0.20	0.40	0.255 (0.000)	0.503	—	0.241	0
0.10	0.20	0.60	0.712 (0.002)	0.709	0.545	0.548	0
0.10	0.20	0.80	1.511 (0.004)	0.911	0.864	0.865	0
0.10	0.20	1.00	2.739 (0.007)	1.110	1.163	1.164	0
0.10	0.20	1.20	4.480 (0.010)	1.308	1.447	1.447	0
0.10	0.20	1.50	8.238 (0.017)	1.603	1.854	1.855	19
0.10	0.20	2.00	18.222 (0.032)	2.089	2.504	2.505	119
0.10	0.20	3.00	56.409 (0.084)	3.044	3.744	3.748	319
0.20	0.05	0.40	0.271 (0.000)	0.514	0.248	0.254	0
0.20	0.05	0.60	0.671 (0.000)	0.695	0.524	0.526	0
0.20	0.05	0.80	1.289 (0.000)	0.864	0.790	0.792	0
0.20	0.05	1.00	2.151 (0.000)	1.025	1.033	1.037	0
0.20	0.05	1.20	3.286 (0.001)	1.180	1.262	1.264	31
0.20	0.05	1.50	5.567 (0.001)	1.407	1.584	1.585	127
0.20	0.05	2.00	11.217 (0.003)	1.777	2.089	2.090	287
0.20	0.05	3.00	31.893 (0.007)	2.517	3.066	3.066	607
0.20	0.10	0.40	0.249 (0.000)	0.499	—	0.236	0
0.20	0.10	0.60	0.661 (0.001)	0.692	0.511	0.520	0
0.20	0.10	0.80	1.340 (0.001)	0.875	0.805	0.810	0
0.20	0.10	1.00	2.331 (0.002)	1.052	1.074	1.078	0
0.20	0.10	1.20	3.674 (0.003)	1.225	1.326	1.329	0
0.20	0.10	1.50	6.426 (0.005)	1.476	1.679	1.681	59
0.20	0.10	2.00	13.259 (0.011)	1.879	2.226	2.226	159
0.20	0.10	3.00	37.225 (0.029)	2.650	3.241	3.239	359
0.20	0.20	0.40	0.282 (0.000)	0.521	0.248	0.264	0
0.20	0.20	0.60	0.750 (0.001)	0.721	0.553	0.567	0
0.20	0.20	0.80	1.535 (0.003)	0.916	0.864	0.872	0
0.20	0.20	1.00	2.700 (0.006)	1.105	1.150	1.156	0
0.20	0.20	1.20	4.306 (0.009)	1.291	1.419	1.423	0
0.20	0.20	1.50	7.655 (0.015)	1.564	1.800	1.802	19
0.20	0.20	2.00	16.144 (0.030)	2.006	2.395	2.396	119
0.20	0.20	3.00	46.330 (0.078)	2.851	3.501	3.499	319

Table S1. Collapse runs (q=1, κ=24).

S5. Runs without a trapped surface at the base resolution

Table S2 lists the seven runs of Table S1 in which no trapped surface was found at the base resolution: the cause of the stop, the last row reached, the largest σ on the last good row and the outcome of refinement by factors of 2 and 4 (the radius of the first trapped point, or none).

A	w	R_w	m_{ext}	r_h	Stop	u_{end}	σ_{max}	$h/2$	$h/4$
0.05	0.10	0.40	0.198	0.191	non-finite values	0.58	2.26	none	0.184
0.05	0.20	0.40	0.324	0.298	non-finite values	0.58	1.54	none	0.233
0.05	0.20	2.00	48.383	3.552	non-finite values	0.37	2.79	2.547	2.534
0.05	0.20	3.00	110.863	4.735	non-finite values	0.29	2.70	3.834	3.821
0.10	0.10	0.40	0.203	0.196	non-finite values	0.54	1.86	none	none
0.10	0.20	0.40	0.255	0.241	non-finite values	0.56	1.94	0.234	0.234
0.20	0.10	0.40	0.249	0.236	non-finite values	0.46	2.10	0.229	0.229

Table S2. Runs without a trapped surface at the base resolution (q=1).

S6. Exteriors of other vacuum depth

Table S3 lists the runs of Section 5.5 (A=0.1): vacuum depth q, wall width w, core radius R_w, exterior mass in units of $m_c=(1+q)R_{dS}$, the horizon radius of the exterior (none above the Nariai mass for q<0), the radius of the first trapped point if one was found, the number of anti-trapped core points on the last row and the cause of the stop.

q	w	R_w	m_{ext}/m_c	r_h	r (first trapped)	Anti-trapped points	Stop
−0.5	0.05	0.60	0.46	0.238	—	0	non-finite values
−0.5	0.05	0.80	0.86	0.485	—	0	non-finite values
−0.5	0.05	1.00	1.36	none	—	0	non-finite values
−0.5	0.05	1.20	1.95	none	—	31	non-finite values
−0.5	0.05	1.50	2.93	none	—	127	non-finite values
−0.5	0.05	2.00	4.40	none	—	287	non-finite values
−0.5	0.05	3.00	3.48	none	—	607	non-finite values
−0.5	0.10	0.60	0.55	0.289	—	0	non-finite values
−0.5	0.10	0.80	1.07	0.727	—	0	non-finite values
−0.5	0.10	1.00	1.78	none	—	0	non-finite values
−0.5	0.10	1.20	2.70	none	—	0	non-finite values
−0.5	0.10	1.50	4.46	none	—	59	non-finite values
−0.5	0.10	2.00	8.33	none	—	159	non-finite values
−0.5	0.10	3.00	18.31	none	—	359	non-finite values
0.0	0.05	0.60	0.34	0.339	—	0	non-finite values
0.0	0.05	0.80	0.68	0.684	0.680	0	non-finite values
0.0	0.05	1.00	1.18	1.180	1.177	0	non-finite values
0.0	0.05	1.20	1.84	1.841	1.834	31	non-finite values
0.0	0.05	1.50	3.15	3.154	—	127	non-finite values
0.0	0.05	2.00	6.21	6.213	—	287	non-finite values
0.0	0.05	3.00	15.29	15.290	—	607	non-finite values
0.0	0.10	0.60	0.38	0.383	—	0	non-finite values
0.0	0.10	0.80	0.79	0.786	—	0	non-finite values
0.0	0.10	1.00	1.38	1.383	1.376	0	non-finite values
0.0	0.10	1.20	2.20	2.201	2.187	0	non-finite values
0.0	0.10	1.50	3.90	3.897	—	59	non-finite values
0.0	0.10	2.00	8.13	8.132	—	159	non-finite values
0.0	0.10	3.00	22.58	22.580	—	359	non-finite values
2.0	0.05	0.60	0.26	0.507	0.501	0	non-finite values
2.0	0.05	0.80	0.57	0.774	0.771	0	non-finite values
2.0	0.05	1.00	1.06	1.024	1.021	0	non-finite values
2.0	0.05	1.20	1.76	1.263	1.261	31	non-finite values
2.0	0.05	1.50	3.30	1.607	1.606	127	non-finite values
2.0	0.05	2.00	7.42	2.158	2.157	287	non-finite values
2.0	0.05	3.00	23.19	3.213	3.213	607	non-finite values
2.0	0.10	0.60	0.27	0.519	0.513	0	non-finite values
2.0	0.10	0.80	0.59	0.791	0.787	0	non-finite values
2.0	0.10	1.00	1.11	1.046	1.042	0	non-finite values
2.0	0.10	1.20	1.86	1.290	1.288	0	non-finite values
2.0	0.10	1.50	3.51	1.645	1.644	59	non-finite values
2.0	0.10	2.00	7.98	2.215	2.216	159	non-finite values
2.0	0.10	3.00	25.41	3.316	3.316	359	non-finite values

Table S3. Collapse runs with other exterior vacua (A=0.1).

S7. Convergence of the constraints and of the null expansions

Fig. S1 shows the RMS residuals of C_{vv} and C_{uu} against the grid step for the run with a core inside its de Sitter radius (R_w=1, region 0.05≤u≤0.28, u+0.4≤v≤3) and for the run with a core outside it (R_w=2, region 0.05≤u≤0.15, u+0.4≤v≤4), and the maximum difference of r_u and r_v at u=0.2 (first run) and u=0.12 (second) between successive resolutions; Table S4 gives the numbers, absolute and relative to the RMS of the largest term of each constraint.

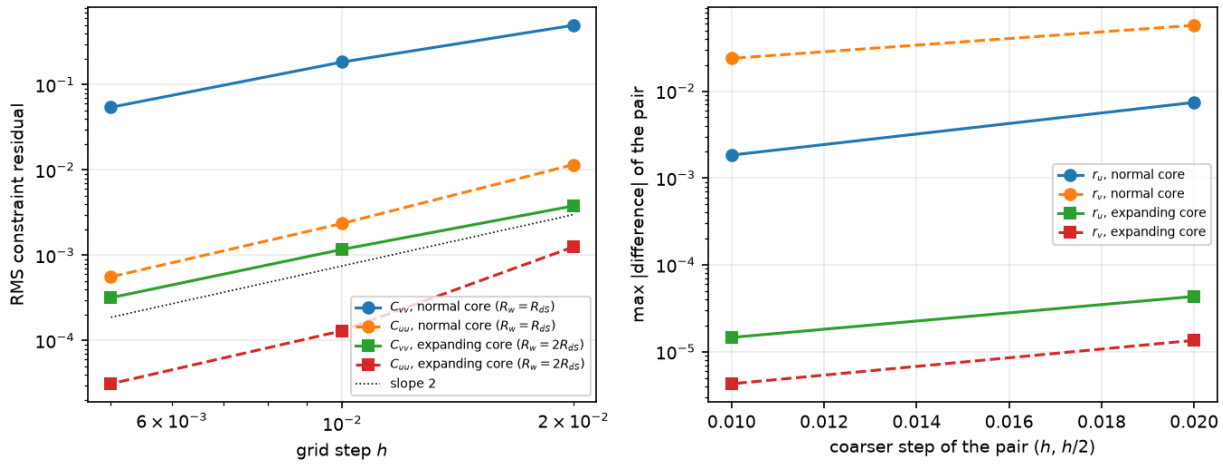


Fig. S1. (a) RMS residuals of the constraints against the grid step. (b) Maximum difference of the null expansions between successive resolutions.

Run	h	Points	RMS C_{vv}	relative	RMS C_{uu}	relative
R_w=1	0.020	1,470	4.97×10^{-1}	34.0%	1.16×10^{-2}	1.86%
R_w=1	0.010	5,612	1.84×10^{-1}	11.6%	2.36×10^{-3}	0.36%
R_w=1	0.005	22,425	5.44×10^{-2}	3.3%	5.59×10^{-4}	0.08%
R_w=2	0.020	1,053	3.79×10^{-3}	2.7%	1.25×10^{-3}	0.09%
R_w=2	0.010	3,505	1.17×10^{-3}	0.8%	1.30×10^{-4}	0.01%
R_w=2	0.005	14,010	3.18×10^{-4}	0.2%	3.12×10^{-5}	0.00%

Table S4. RMS constraint residuals of the dynamical solutions.

S8. Independent audit of the junction branches, critical mass, horizons and masses

independent_audit.py re-derives the junction conditions symbolically from the extrinsic curvatures and does not use the closed forms of thinwall.py except for the final comparison. Eliminating Γ_{out} from $\Gamma_{in}-\Gamma_{out}=\sigma R/2$ and $\Gamma_{out}^2-\Gamma_{in}^2=f_{out}-fi_n$ gives Eq. (7), and the roots of $U^2=0$ in m reproduce Eq. (8): the maximum deviation over 200 random (R,σ,q) is 0.0, and the two branches meet at $R=R_{dS}$ at $m=(\Delta/3-\sigma^2/4)R_{dS}^3$. The maximum of $m_+(R)$ was recomputed with 120-digit arithmetic (Table S5); at the model's tension $m_c-2 = -1.223 \times 10^{-60}$, the series gives -1.223×10^{-60}, and the maximum lies 1.02×10^{-61} below R_{dS}.

q	$\hat{\sigma}$	m_c (120 digits)	Series, Eq. (9)	R at the maximum
0.5	0.01	1.499986111	1.499986111	0.999998
0.5	0.10	1.498612003	1.498611111	0.999753
0.5	0.30	1.487572212	1.487500000	0.997758
0.5	0.90	1.393323869	1.387500000	0.978518
1.0	0.01	1.999983333	1.999983333	0.999999
1.0	0.10	1.998333970	1.998333333	0.999861
1.0	0.30	1.985051698	1.985000000	0.998737
1.0	0.90	1.869269063	1.865000000	0.987669
2.0	0.01	2.999980556	2.999980556	0.999999
2.0	0.10	2.998055899	2.998055556	0.999938
2.0	0.30	2.982527879	2.982500000	0.999439
2.0	0.90	2.844824346	2.842500000	0.994545

Table S5. Critical mass recomputed with 120-digit arithmetic against the small-tension series of Eq. (9) (R_{dS}=1).

Horizon roots at $m=m_c=(1+q)R_{dS}$ and mass definitions are in Table S6. For $q>0$ the single exterior horizon is at R_{dS}; for $q=0$ it is the Schwarzschild radius; for $q=-0.5$ the roots are the black-hole horizon at $0.618R_{dS}$ and the cosmological horizon at R_{dS}, so the equality of horizon areas is with the cosmological horizon there. The quasi-local Misner–Sharp mass at the wall at $R=R_{dS}$ equals R_{dS} on both sides for every q, while the exterior parameter is $(1+q)R_{dS}$. Dimensional restoration used independently chosen constants (c, G, M_{odot}, Planck 2018 H_0 and Ω_Λ): $R_{dS}=5.3384$ Gpc, and the critical masses agree with thinwall.json to 2×10^{-16}.

q	m (parameter)	m_{MS} at the wall	Horizon roots r/R_{dS}	M_c (M_{odot})	M_{MS} (M_{odot})
−0.50	0.50	1.00	0.618, 1.000	2.79×10^{22}	5.58×10^{22}
0.00	1.00	1.00	1.000	5.58×10^{22}	5.58×10^{22}
0.50	1.50	1.00	1.000	8.37×10^{22}	5.58×10^{22}
1.00	2.00	1.00	1.000	1.12×10^{23}	5.58×10^{22}
2.00	3.00	1.00	1.000	1.67×10^{23}	5.58×10^{22}

Table S6. Horizons and masses at $m=m_c$ ($R_{dS}=1$, $8\pi G=1$): exterior parameter, quasi-local mass at the wall at $R=R_{dS}$ and horizon roots; critical mass in solar masses with the quasi-local value.

S9. Wall-thickness extrapolation and causal diagnostic

Table S7 gives, for each amplitude A (thickness and tension scale together), the wall thickness and tension, the median and maximum absolute deviation of the first trapped radius from the horizon radius of the measured mass, the resolution floor, the shift of the critical mass at the simulated tension and the smallest m_{ext}/m_c among the expanding cores with the limit and with the finite-tension mass (wall_extrapolation.py). Table S8 gives the causal diagnostic for the two runs of Fig. 5 (causal_diagnostic.py).

A	Thickness/R_{dS}	$\hat{\sigma}$	median |dev|	max |dev|	floor h/r_h	m_c shift	min m/m_c (limit)	min m/m_c (finite $\hat{\sigma}$)
0.05	0.0144	0.231	0.75%	4.1%	1.0%	−0.44%	1.84	1.85
0.1	0.0289	0.462	0.20%	1.1%	1.0%	−1.76%	1.78	1.82
0.2	0.0577	0.924	0.47%	6.2%	1.1%	−6.87%	1.64	1.76

Table S7. Wall-thickness dependence of the numerical outputs (normal-core runs with a trapped surface).

Run	Rows	u_{last}	Anti-trapped cells (core / wall / beyond wall)	Core anti-trapped with exterior in causal past	Wall drift
R_w = 0.8 R_dS	25	0.48	0 / 0 / 0	0	0.095
R_w = 2.0 R_dS	16	0.30	2,981 / 234 / 479	0	0.115

Table S8. Causal diagnostic on the evolved domain.